

Observed NumStability use by expected tier

Twelve alternative measures of realized library use, with identical time, source-line, and observed-token gain outcomes. Every point is a task mean over three N/L pairs (38 tasks; 114 pairs). The archive's 0-3 score is not an axis in this report.

● T1 (expected moderate use) ● T2 (expected strongest use) ● T3 (expected little/no use)

Task-level medians

Tier	Tasks	Distinct names	Occurrences	Audited deps	Time gain
T1	12	6.3	11.3	23.2	+57.6%
T2	3	11.0	18.3	95.3	+75.3%
T3	23	0.0	0.0	0.0	+4.6%

Critical limitation: tier and runner campaign are perfectly confounded in this cohort. T1/T2 came from r11; T3 came from r09/r10. The plotted pooled correlations are descriptive and cannot isolate tier, library use, task difficulty, or runner effects.

Positive gain = (mean N - mean L) / mean N. Qualified names count source text; audited dependencies count the accepted theorem's recorded Lean dependency closure. Token totals are provider-observed. Sources: archived attempt_usage.csv, task_usage.csv, pair_usage.csv and metadata/manifest.json. No task is excluded from this 38-task analysis cohort.

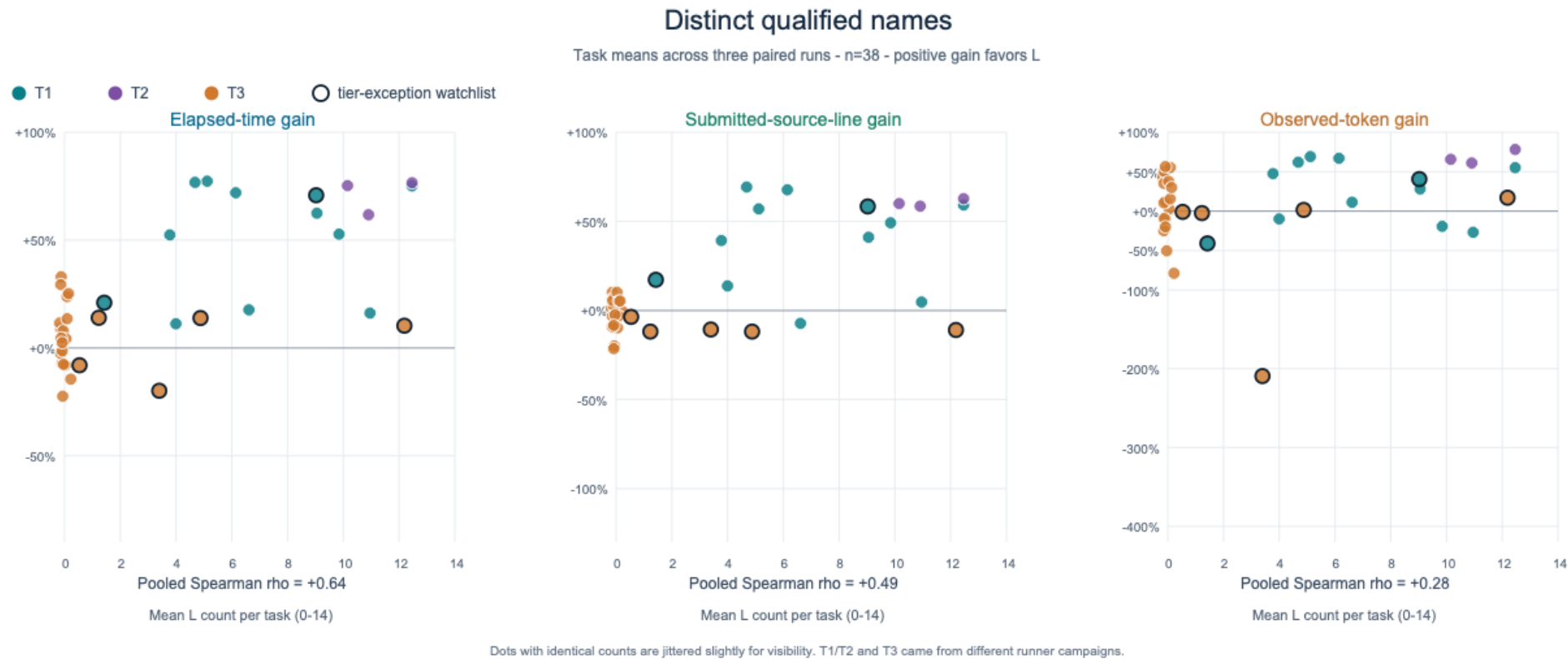
Tasks that challenge their expected tier

Diagnostic rule: flag T1 when at most one of three L runs uses a prelisted substantial result; flag T2 if fewer than three do; flag T3 when any L run has audited library use. This intentionally sensitive rule has seven flags, separated by materiality below. It is a post-run diagnostic, not a change to the task corpus.

Task	Signal	Use runs	Same-paper other tier	What does not fit
P15-T3	Clear	3/3 audited	P15-T1; P15-T2	All three runs use many qualified names and audited dependencies.
P17-T1	Clear	1/3 audited; 0/3 substantive	P17-T3	Two runs have no audited use; no run uses a prelisted substantial result.
P02-T3	Episodic	1/3 audited	P02-T1	One of three runs uses 10 qualified names and 22 audited dependencies.
P40-T3	Episodic	2/3 audited	None	Two runs use library names; one has five direct audited dependencies.
P33-T1	Partial	3/3 audited; 1/3 substantive	None	Audited use in all runs, but prelisted substantial use in only one.
P20-T3	Trace	1/3 audited	None	One run has four qualified names and four audited dependencies.
P35-T3	Trace	2/3 audited	None	Two runs show only a single qualified name and small dependency closures.

The strongest departures are P15-T3 and P17-T1. P20-T3 and P35-T3 are trace-level departures, not evidence of broad reusable coverage. P28-T1 and P29-T1 have zero task-listed candidate hits but substantial audited use in all three runs; this is a candidate-list limitation, not a tier failure.

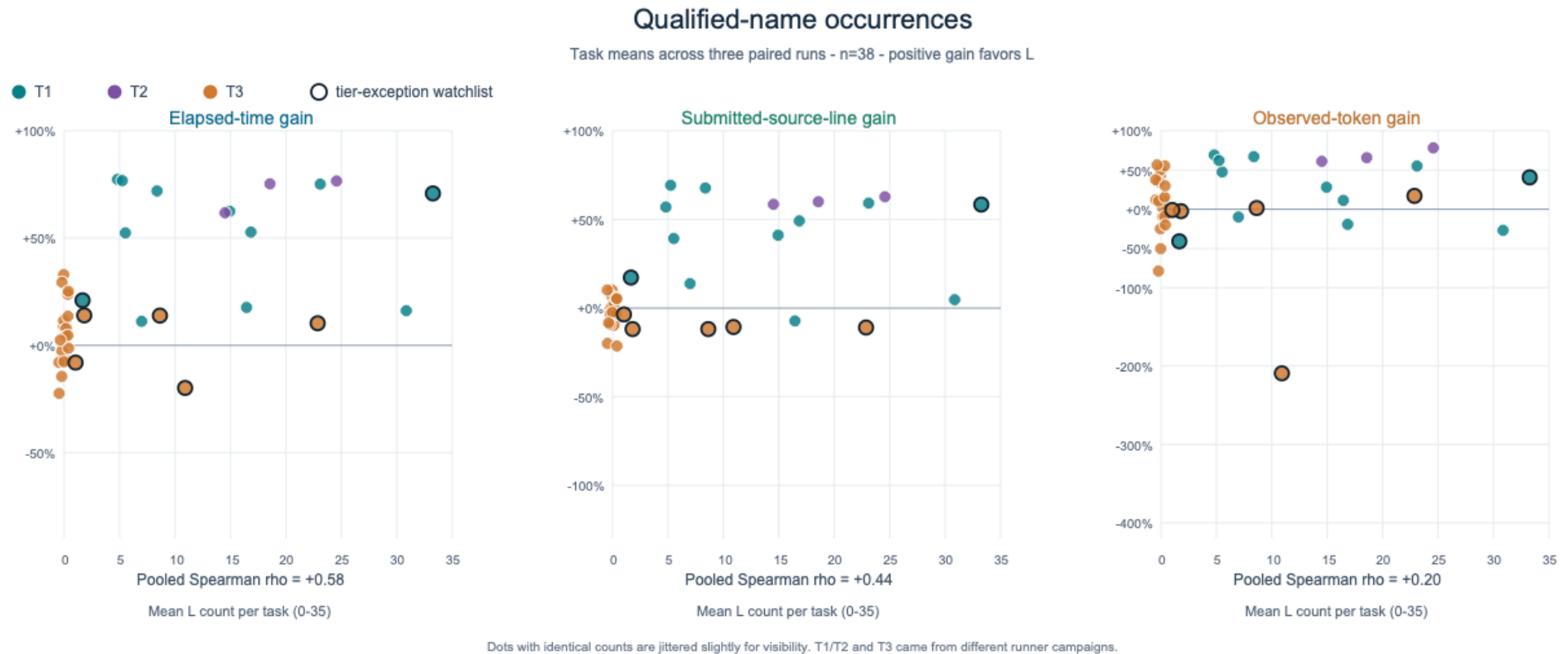
Metric 1/12: Distinct qualified names



Definition: Distinct NumStability.* names written outside imports, comments and strings.

The pooled relationship mixes expected tier with runner campaign. See the source CSV for task means and all within-campaign correlations.

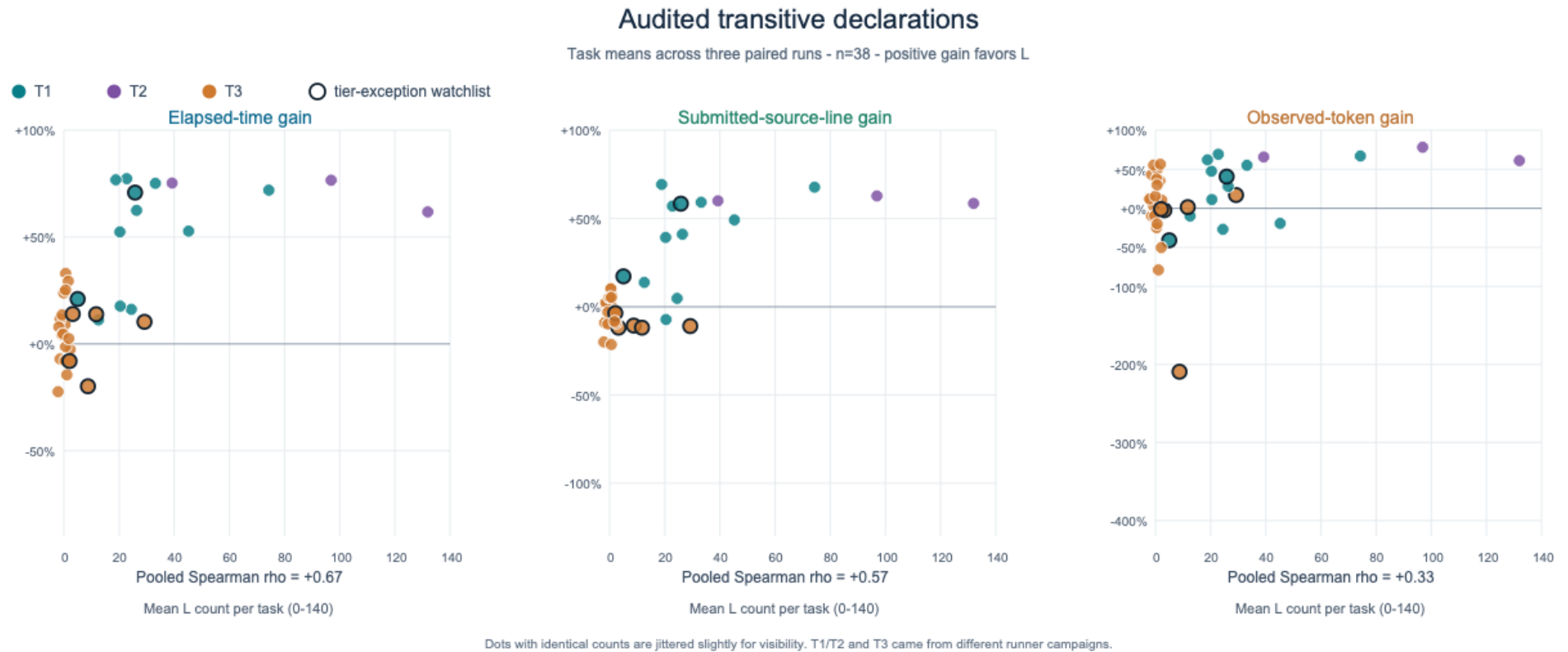
Metric 2/12: Qualified-name occurrences



Definition: Every NumStability.* occurrence written outside imports, comments and strings.

The pooled relationship mixes expected tier with runner campaign. See the source CSV for task means and all within-campaign correlations.

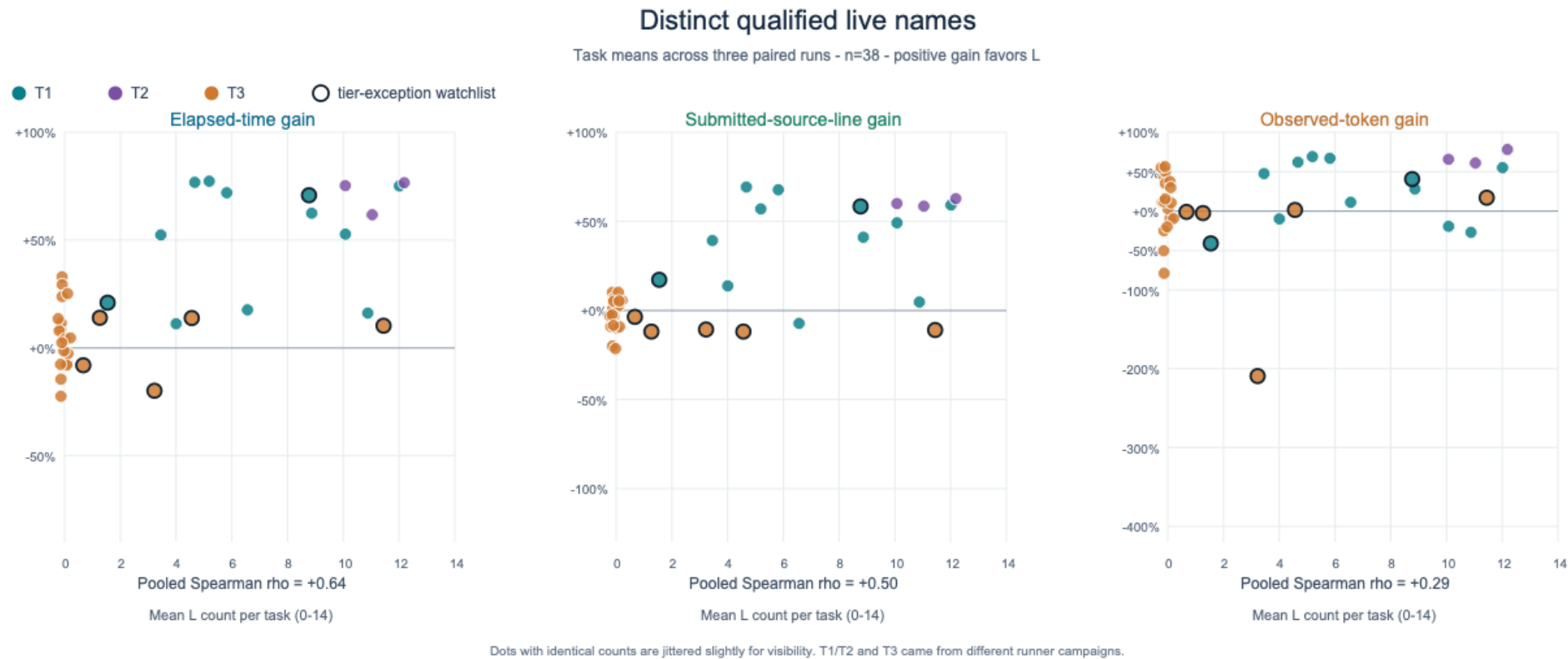
Metric 3/12: Audited transitive declarations



Definition: All NumStability dependencies in the accepted theorem's audited dependency closure.

The pooled relationship mixes expected tier with runner campaign. See the source CSV for task means and all within-campaign correlations.

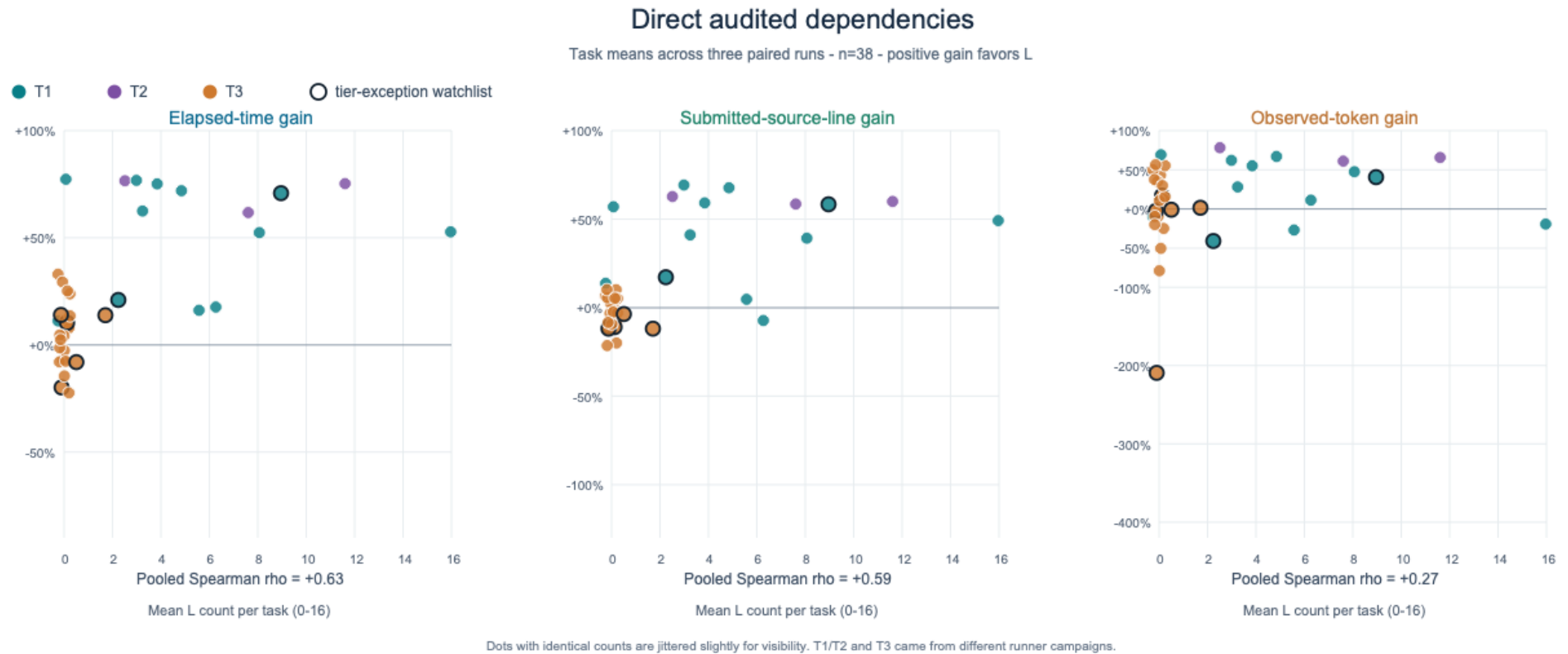
Metric 4/12: Distinct qualified live names



Definition: Written qualified names also matched to a name in the audited closure.

The pooled relationship mixes expected tier with runner campaign. See the source CSV for task means and all within-campaign correlations.

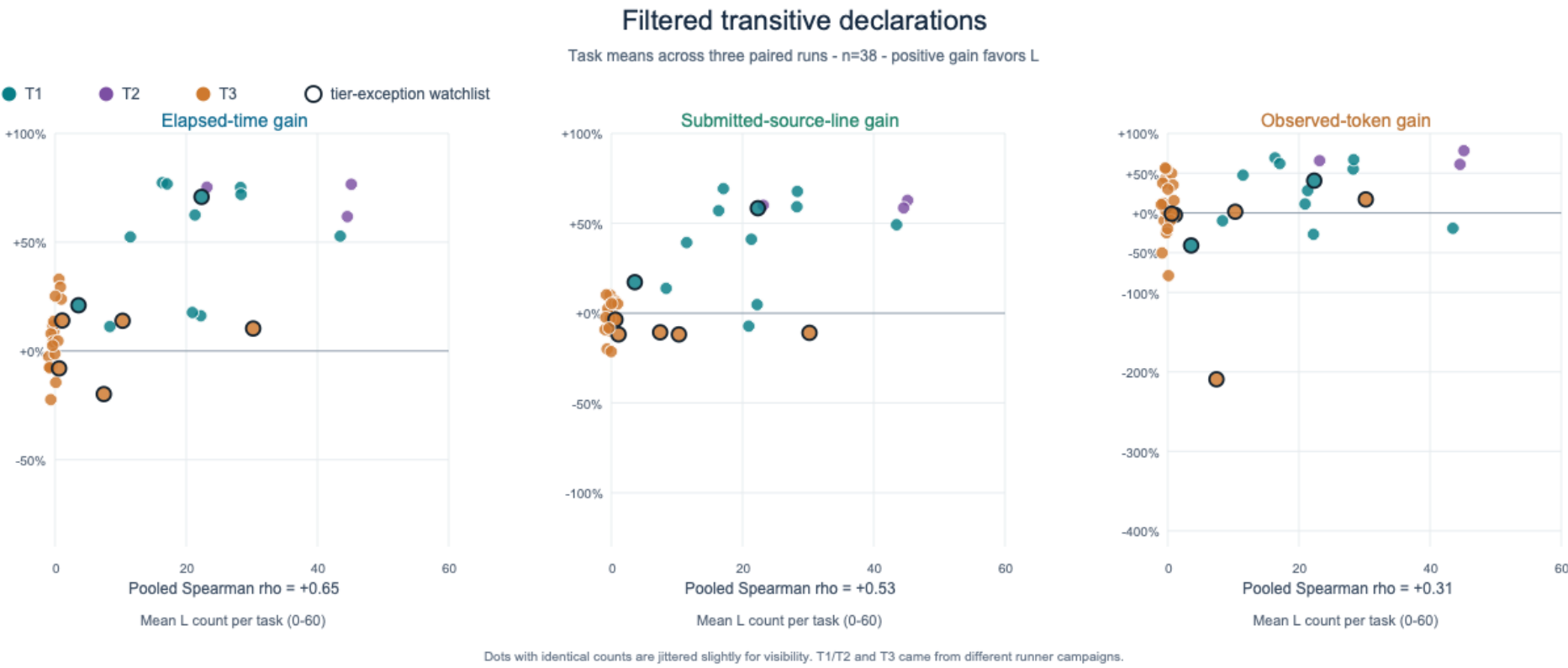
Metric 5/12: Direct audited dependencies



Definition: NumStability declarations at dependency-graph distance one from the accepted theorem.

The pooled relationship mixes expected tier with runner campaign. See the source CSV for task means and all within-campaign correlations.

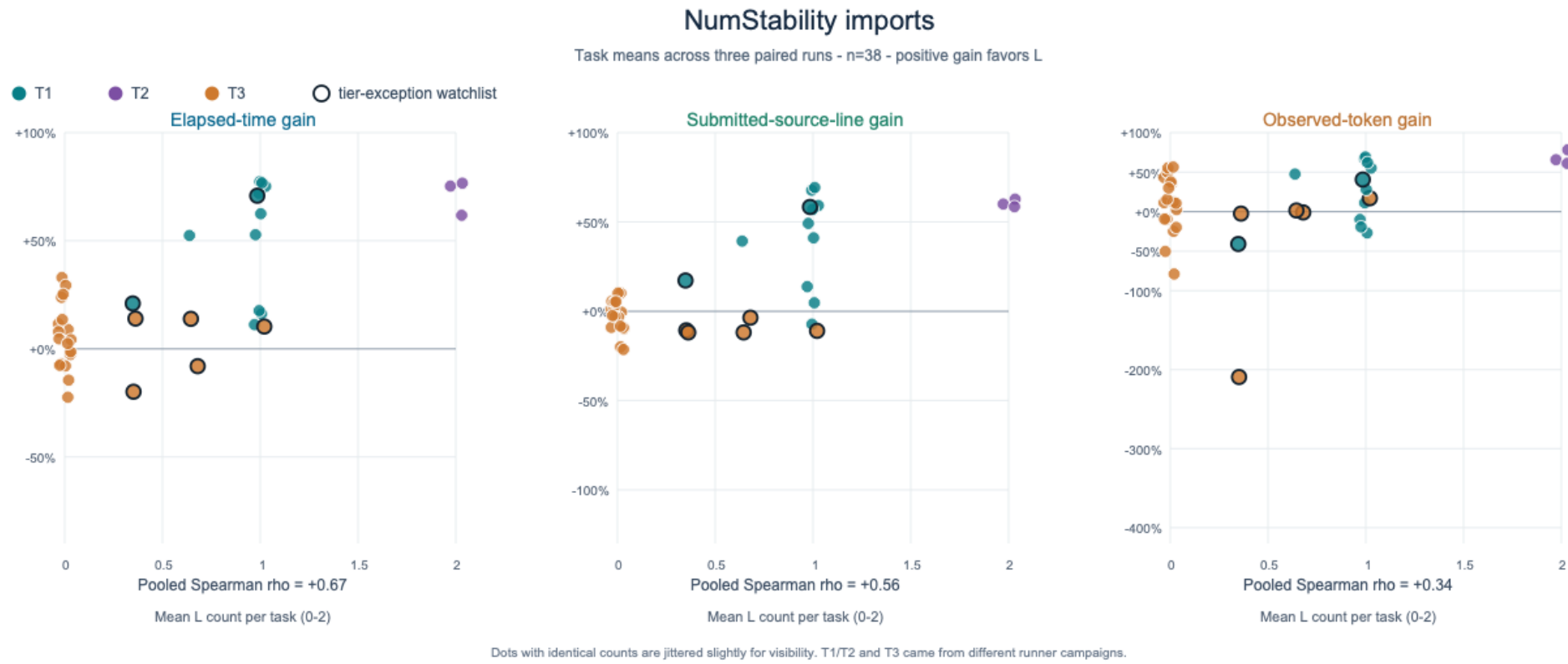
Metric 6/12: Filtered transitive declarations



Definition: Transitive declarations after the archive's limited generated-name filter.

The limited filter omits some generated-name patterns but does not certify that every remaining dependency belongs to a public API.

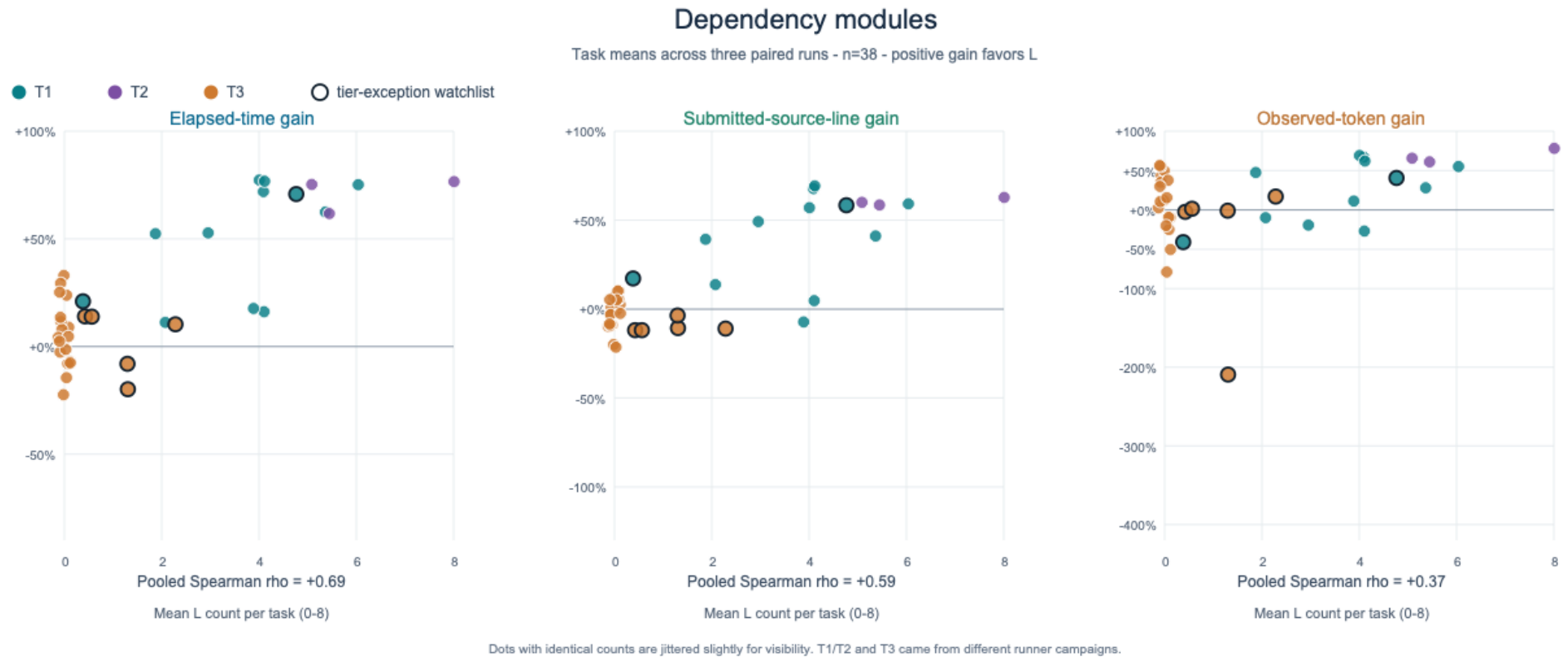
Metric 7/12: NumStability imports



Definition: Distinct NumStability import lines in the accepted source.

An import provides access; it does not establish that a declaration was used in the accepted theorem.

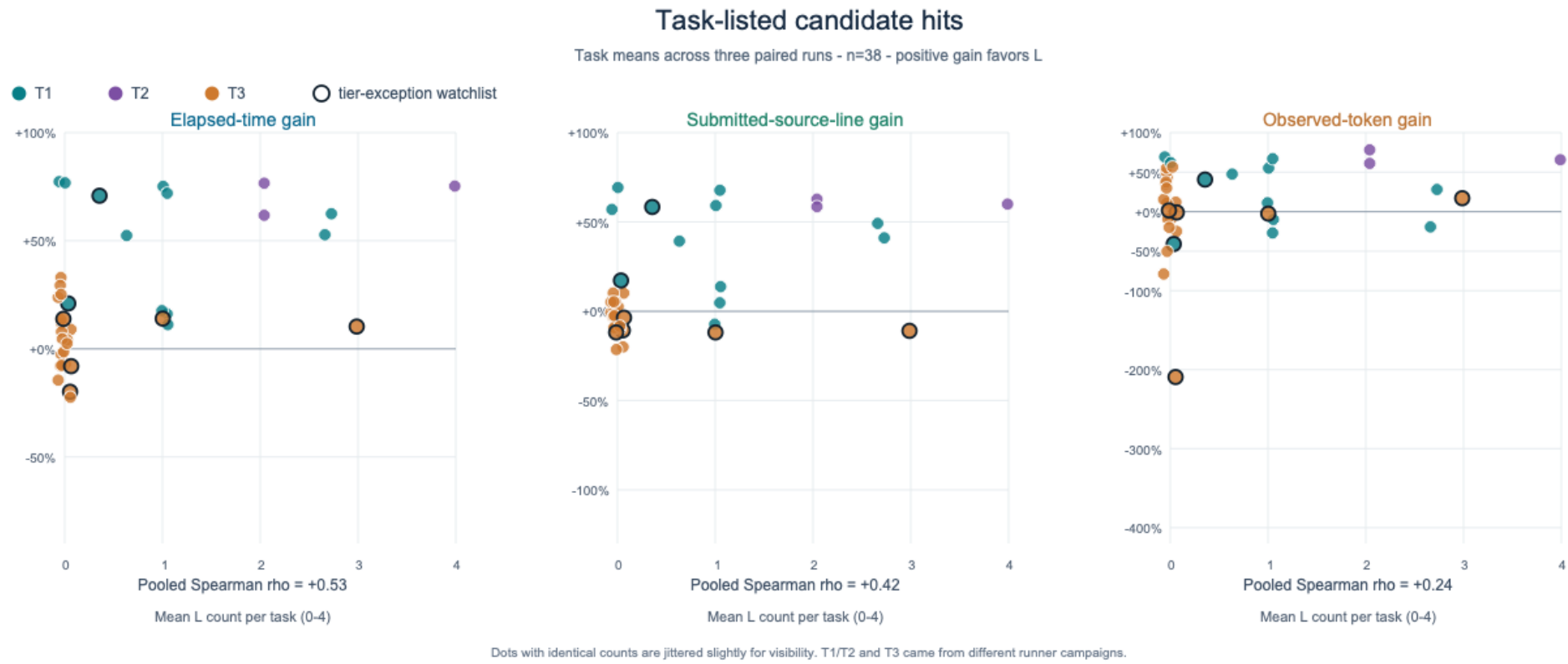
Metric 8/12: Dependency modules



Definition: Distinct NumStability modules owning audited dependency declarations.

The pooled relationship mixes expected tier with runner campaign. See the source CSV for task means and all within-campaign correlations.

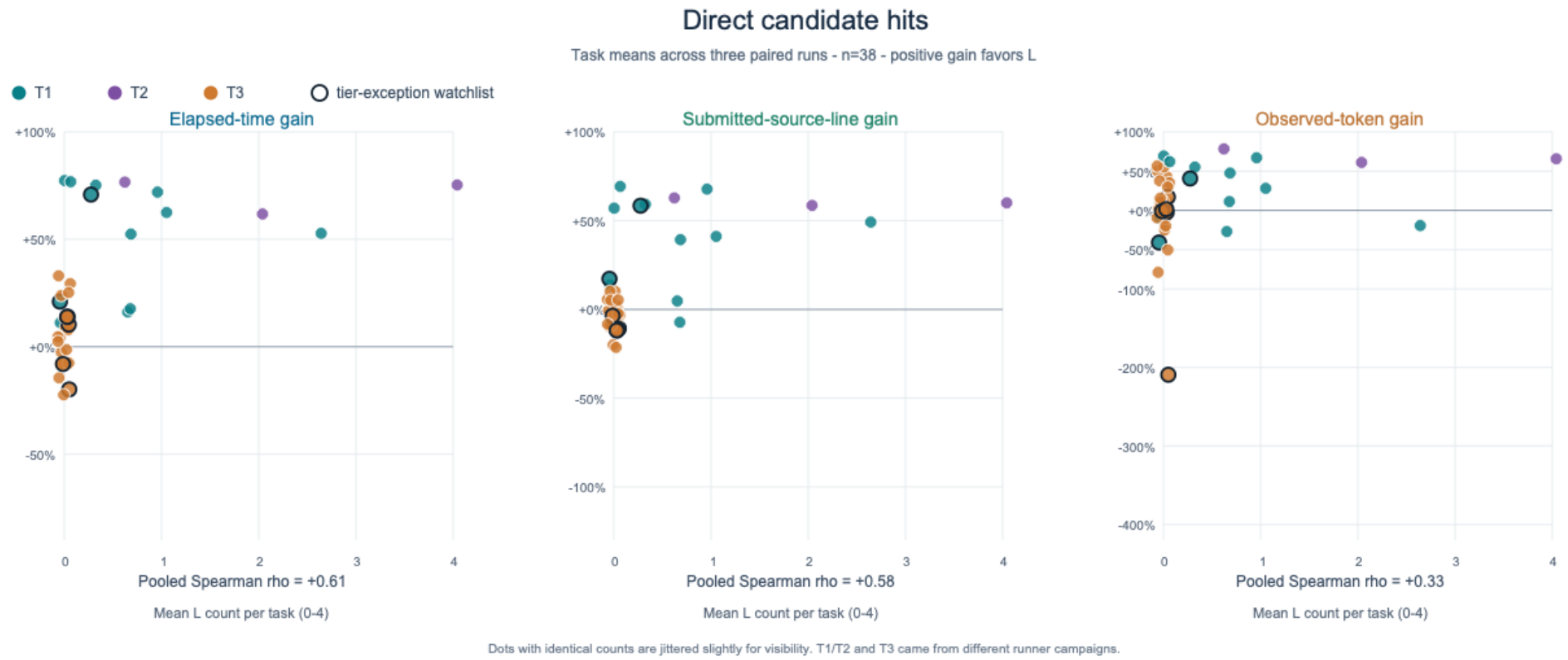
Metric 9/12: Task-listed candidate hits



Definition: Task-specific candidate names present anywhere in the audited dependency closure.

This measure uses task-specific candidate or substantial-result classifications. T3 has no predesignated substantial result, so substantial-hit zeros there are partly by construction.

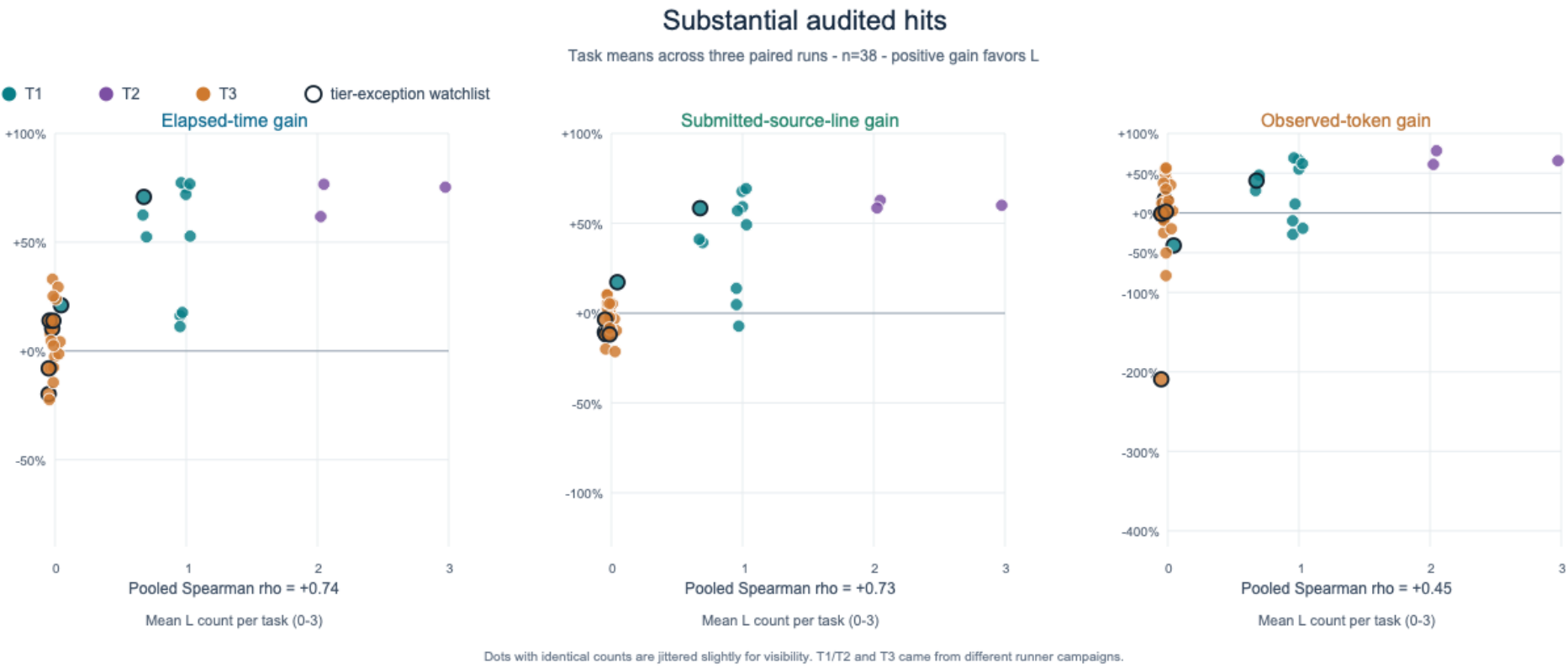
Metric 10/12: Direct candidate hits



Definition: Task-specific candidate names at dependency-graph distance one.

This measure uses task-specific candidate or substantial-result classifications. T3 has no predesignated substantial result, so substantial-hit zeros there are partly by construction.

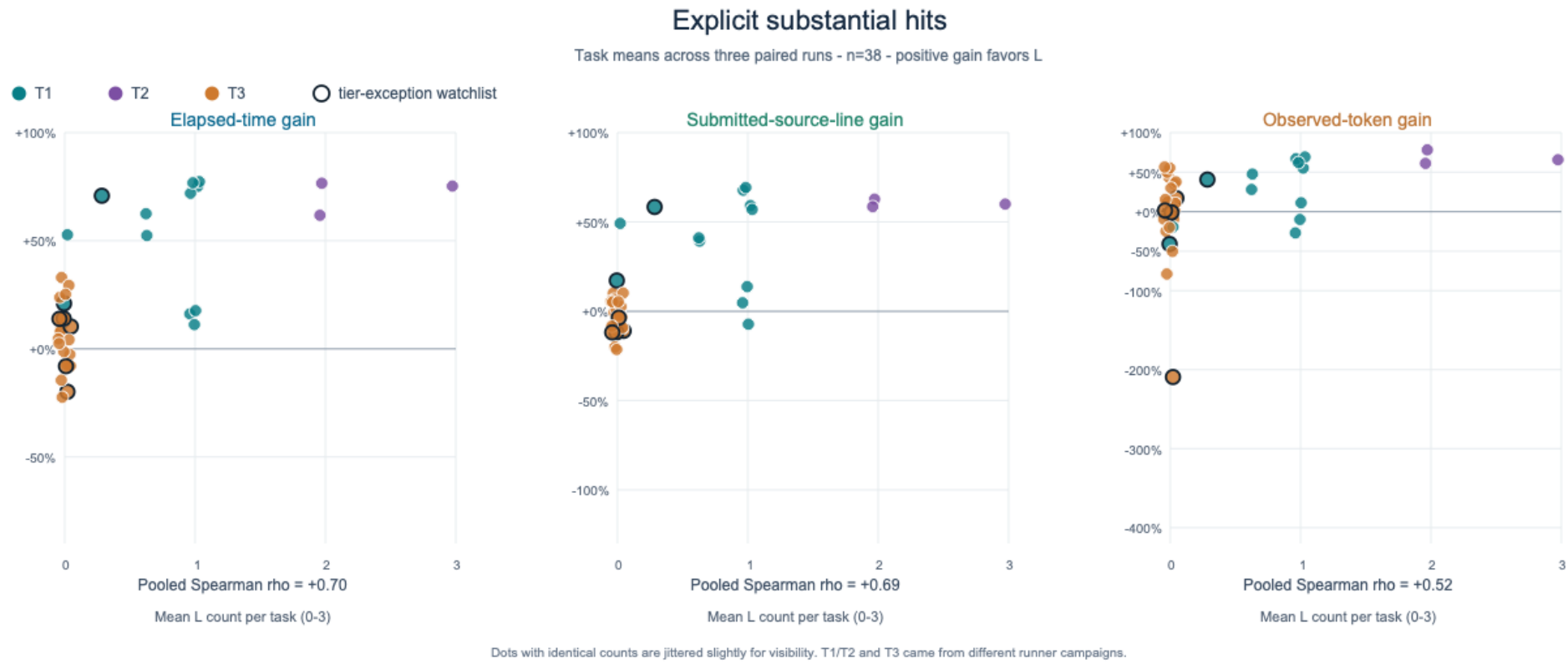
Metric 11/12: Substantial audited hits



Definition: Task-relevant substantial declarations in the audited closure.

This measure uses task-specific candidate or substantial-result classifications. T3 has no predesignated substantial result, so substantial-hit zeros there are partly by construction.

Metric 12/12: Explicit substantial hits



Definition: Task-relevant substantial declarations named directly in source.

This measure uses task-specific candidate or substantial-result classifications. T3 has no predesignated substantial result, so substantial-hit zeros there are partly by construction.

Source and interpretation notes

Inputs: `paper_benchmark/highambench/reports/library-usage-analysis/attempt_usage.csv`, `task_usage.csv`, `pair_usage.csv`, `analyze.py` and `semantic_catalog.json`; `paper_benchmark/highambench/metadata/manifest.json`. The accepted-proof hashes and dependency records are preserved in the attempt-level CSV. The source CSV byte hashes are checked by `make_plots.py`, which this builder imports.

Aggregation: each L-side usage count is averaged over three accepted runs per task. Each gain is computed from the three-run condition means, not the mean of three per-pair percentage gains. The full 38-task analysis cohort is retained; P16-T3 had already been excluded from the source cohort because its target was documented false.

Qualified occurrences count repeated textual NumStability.* references outside imports, comments and strings. Distinct qualified names count each such spelling once. Live qualified names additionally match a dependency name. Audited transitive declarations include indirect dependencies and can contain implementation-generated names. Distance-one declarations are the direct graph neighbors of the accepted theorem.

Interpretation: a nonzero lexical or dependency count is evidence of observed use, not an independent measurement of how much helpful mathematics was available. Candidate and substantial counts depend on task-specific catalogs. Counts and performance may co-vary because of problem difficulty, tier design, runner changes, or other factors. These plots do not establish a causal library speedup.

Rebuild: run `python3 make_tier_report.py` from this report directory. The script regenerates the tier-colored SVG/PNG figures, `tier_exceptions.csv` and this PDF. Earlier campaign-colored figures remain separate and are not used in this PDF.